Perfect Competition: Firms and Markets

Market price → output decisions → profit and shutdown → entry, exit, and long-run equilibrium

1. Market structures overview

A business has to decide what to sell, how much to produce, and what price it can charge. The answer depends partly on *market structure*: how many competing sellers there are, whether their products differ, and whether other firms can enter the market. Start with the case in which an individual seller has essentially no control over price. This is *perfect competition*. It is the benchmark against which we will compare other market structures.

> **DEFINITION Market structure**
>
> **Market structure** describes the competitive conditions in a market: the number of firms, whether goods are identical or differentiated, and how easy it is for firms to enter or leave. These conditions affect firms' power over the price they charge.

Structure	Main feature	Firm's pricing position
Perfect competition	Many sellers of identical goods	Price taker
Monopolistic competition	Many sellers of differentiated goods	Some discretion over price
Oligopoly	A few important competing firms	Decisions depend on rivals
Monopoly	One seller without close substitutes; entry barriers	Faces market demand

So what? A bakery selling a distinctive product may lose some customers when its own price rises, but it may not lose all of them. A firm selling an identical, widely available product cannot generally charge more than its competitors. Our diagrams will describe the second case, not every actual business.

2. Conditions for perfect competition

> **DEFINITION Perfect competition and price taking**
>
> In a **perfectly competitive market**, there are many buyers and sellers, firms offer **identical products**, and each buyer and seller is too small to influence the market price. Individual firms are **price takers**: they accept the market price rather than choose it. In the **long run**, firms can enter and exit without significant barriers.

We will use an original hypothetical market for *identical boxes of shipping labels*. Imagine many small printing businesses making the same standardized box. Each firm's output is too small to change the going price. If the market price is $20 per box, buyers can obtain an equivalent box elsewhere for $20. A single small seller that demands $23 will lose its sales to rivals; it need not cut its price to sell one more box at $20.

KEY RESULT The price-taking firm faces a horizontal demand curve

The **market** price is determined by market demand and market supply. A **single competitive firm** can sell its chosen output at that price, so its own demand curve is horizontal at P. Do not confuse the horizontal demand curve *facing one firm* with the downward-sloping demand curve for the *whole market*.

Where does the firm's market price come from?

The price is not decided by the small firm. In the industry as a whole, buyers' demand and sellers' supply interact. An increase in market demand can raise the price facing every existing firm even though no individual firm has gained pricing power.

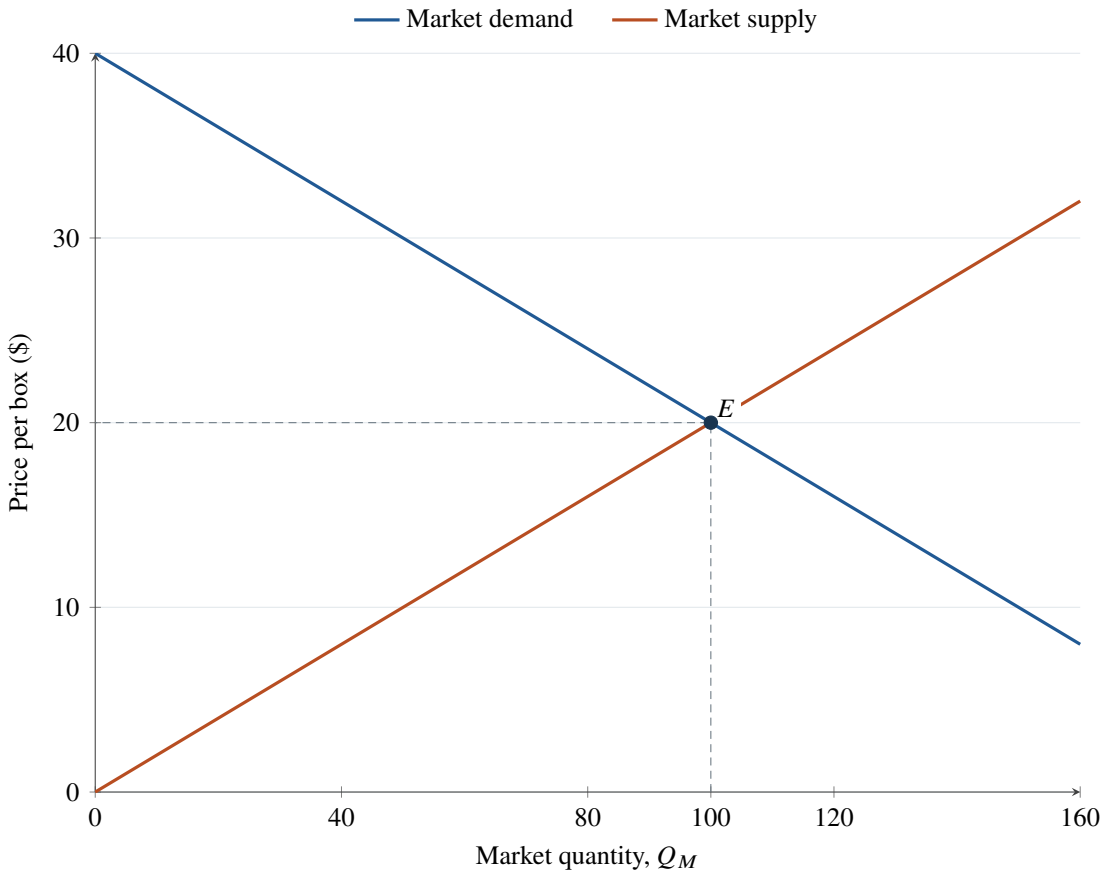


Figure 1. Market demand and supply meet at $20 per box. This market price is taken as given by each small firm. The quantities here refer to the whole market, not one seller.

Read the graph. Market demand and supply intersect at P = $20. A small printing business faces that P. If the market equilibrium changed to $24, its horizontal firm-level demand curve would move to $24. There is no contradiction: the *market* determines price, and the *firm* responds.

3. Marginal revenue equals marginal cost: choosing output

A price-taking firm cannot improve profit simply by raising its price. Its main choice is how many boxes to make. To solve that problem, distinguish revenues and costs, then compare the *next* box's extra revenue with its extra cost.

> **DEFINITION Revenue and economic profit**
>
> **Total revenue** is $TR = P \times Q$. **Average revenue** is $AR = TR/Q$ (when $Q > 0$). **Marginal revenue** is $MR = \Delta TR/\Delta Q$, the extra revenue from selling an additional unit. **Economic profit** is $\pi = TR - TC$, where total cost includes both explicit payments and opportunity costs.

Suppose the market price is \$20. Selling four boxes earns $4 \times 20 = \$80$; selling five earns $5 \times 20 = \$100$. The fifth box adds \$20 of revenue. All the boxes sell for the same price, so the firm earns another \$20 every time it sells one more box.

> **KEY RESULT Revenue of a perfectly competitive firm**
>
> For a price taker, $\boxed{P = AR = MR}$. The firm's horizontal demand line is also its average-revenue and marginal-revenue line. This equality holds *because the firm cannot influence market price by changing its own output*.

> **DEFINITION The marginal decision rule**
>
> **Marginal cost**, MC, is the extra cost of making one more unit. If $MR > MC$, producing that next unit adds more to revenue than cost, so profit rises. If $MR < MC$, making that extra unit reduces profit. At a smooth, interior profit maximum, $\boxed{MR = MC}$ on a rising marginal-cost curve.

Worked example: Choose the output by checking the next unit

The printing firm has a fixed cost of \$70 per day even if it makes no boxes. Its costs of producing additional boxes are shown below. This is a hypothetical *discrete* cost schedule; no calculus is needed.

Q	Total variable cost	Total cost	Total revenue	Profit	MC of last box
0	\$0	\$70	\$0	−\$70	–
1	6	76	20	−56	6
2	15	85	40	−45	9
3	27	97	60	−37	12
4	42	112	80	−32	15
5	60	130	100	−30	18
6	81	151	120	−31	21
7	105	175	140	−35	24
8	133	203	160	−43	28

The fifth box adds \$20 in revenue but only \$18 in cost. The sixth would add \$20 in revenue and \$21 in cost. Thus, **five boxes maximize profit**. The \$30 loss is smaller than the \$70 loss from shutting down.

KEY RESULT The precise rule for tables and curves

For **whole-unit tables**, produce the additional unit while its $MC \leq MR$; stop before the next unit costs more than it brings in. Exact equality is not necessary. For a **smooth cost curve**, choose the relevant rising-MC intersection with MR, *provided that producing is better than shutting down*. The latter condition comes next.

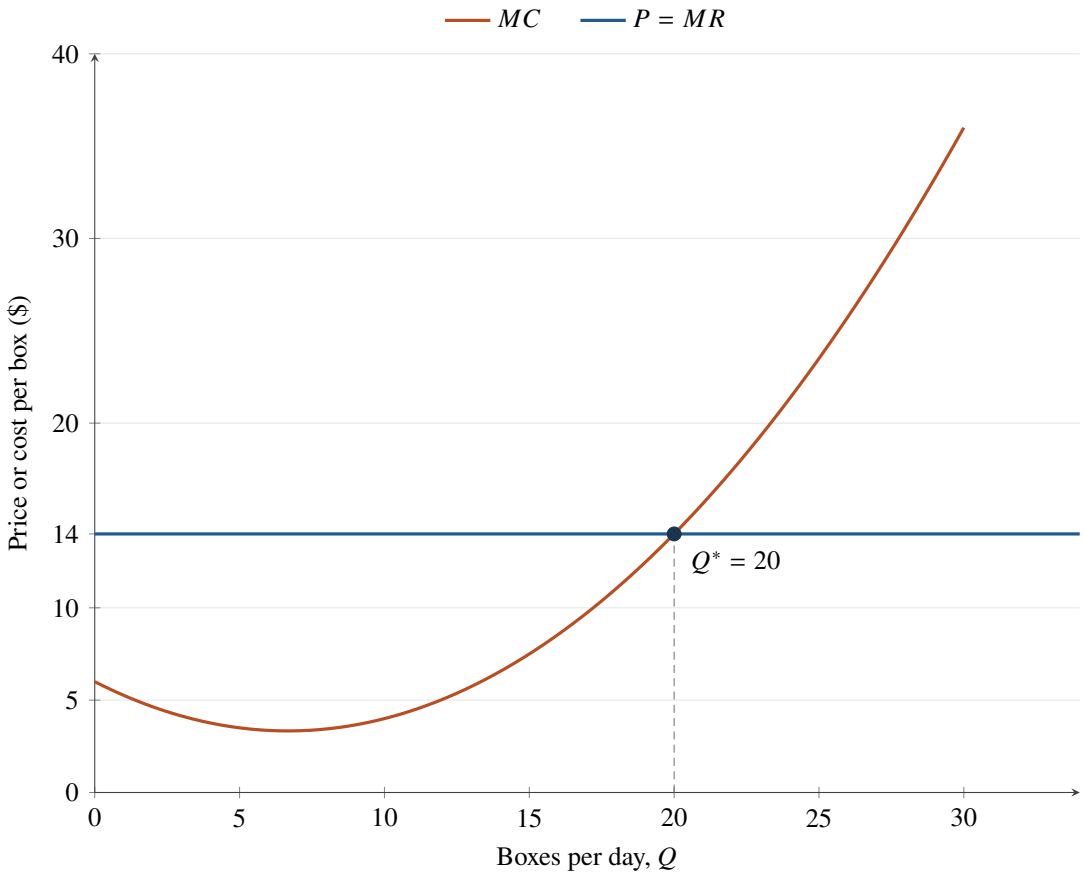


Figure 2. The firm's chosen quantity occurs where the horizontal MR line meets rising MC. Below the chosen output, another unit increases profit; above it, another unit reduces profit.

Read the graph. At quantities just below 20, the extra box costs less than the \$14 selling price. At quantities just above 20, it costs more than \$14. The firm therefore chooses 20 boxes, *if it is worth operating*. Equality of MR and MC identifies output, not whether the firm earns profit.

4. Profit and loss graphically

A firm can choose the best possible output and still lose money. *Profit maximization* does not mean *positive profit*. Once output is chosen by $P = MC$, compare the market price with **average total cost** (ATC), which is cost per unit including fixed and variable costs.

DEFINITION Profit and loss per unit

$ATC = TC/Q$. At any positive output, economic profit is

$$\boxed{\pi = TR - TC = (P - ATC) \times Q.}$$

If $P > ATC$, the firm earns profit. If $P < ATC$, it incurs a loss. If $P = ATC$, economic profit is zero. In a graph, the horizontal distance Q is the width of the profit or loss rectangle; the vertical gap between P and ATC is its height.

Worked example: Profit from a rectangle, not just a formula

Suppose at the output selected by $P = MC$, a firm sells 25 boxes at \$23.50 each. Average total cost at that output is \$14.90. Revenue is $25 \times 23.50 = \$587.50$. Total cost is $25 \times 14.90 = \$372.50$. Economic profit is \$215, exactly $25(23.50 - 14.90)$.

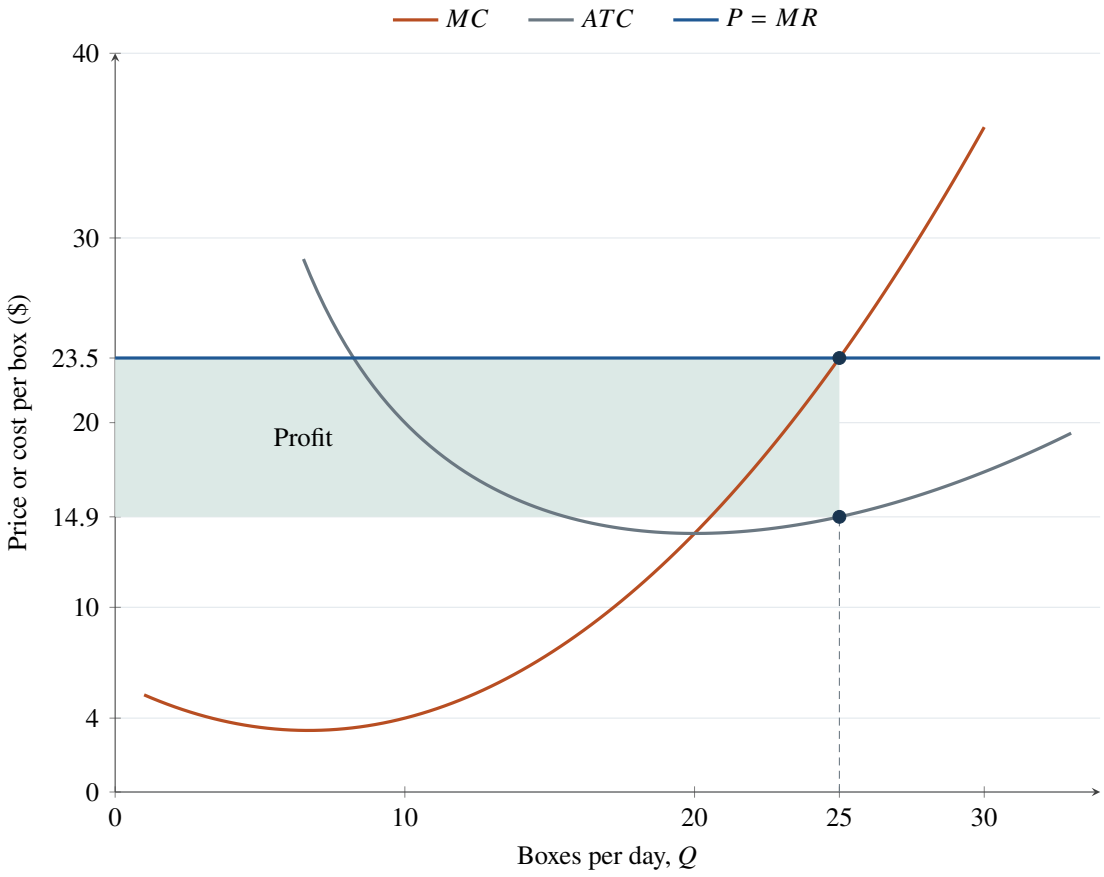


Figure 3. Positive economic profit: at the selected output ($Q = 25$), market price lies above average total cost. The shaded area equals $(P - ATC)Q = \$215$.

Read the graph. Locate the intersection of MR and rising MC, then move straight down to identify 25 boxes. Read *ATC at that same quantity*. The gap between the blue price line and the average-cost curve, multiplied by 25, is total economic profit.

Worked example: A loss can still be the best outcome

Now suppose price is only \$7.50. A firm's chosen output is 15 boxes. At this quantity, $AVC = \$4.50$ and $ATC \approx \$15.17$. The firm loses about $15(15.17 - 7.50) = \$115$, but its \$7.50 price covers all variable cost per box and contributes \$3 per box toward fixed costs. Shutting down would leave the \$160 daily fixed cost unpaid. Thus the firm loses *less* by producing.

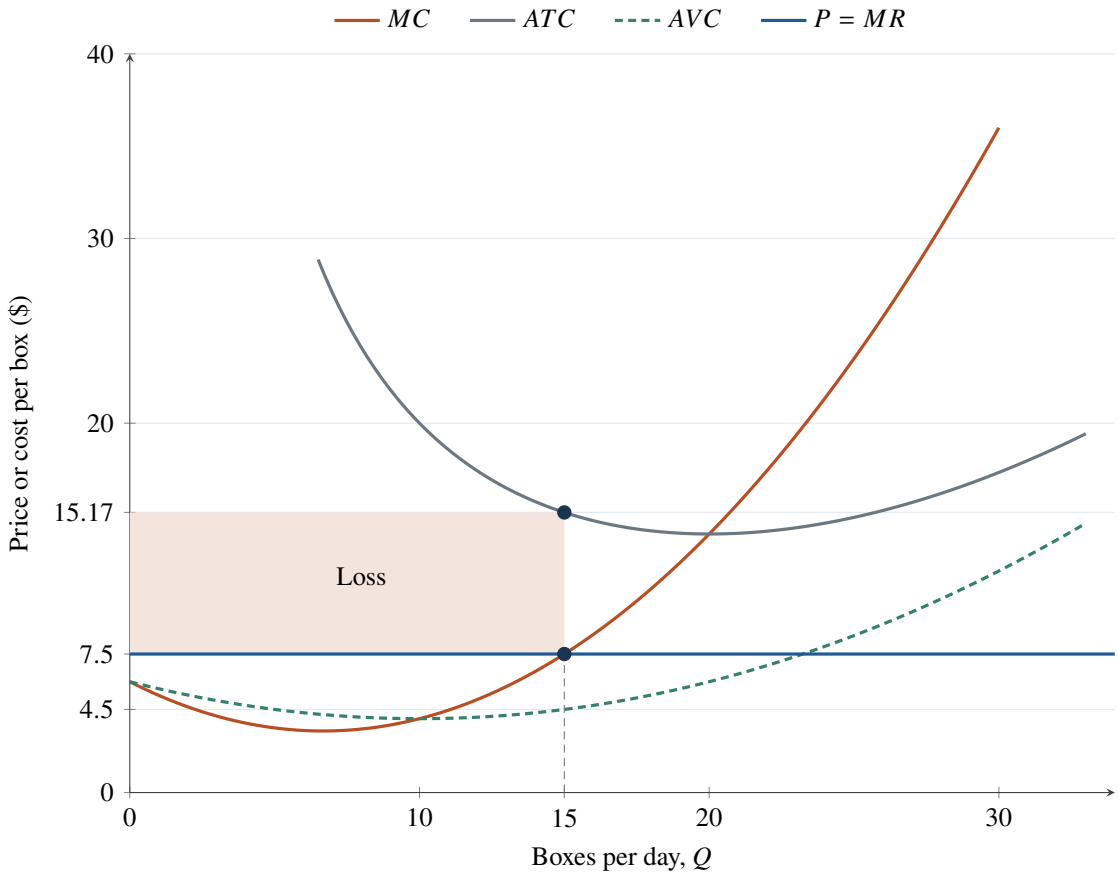


Figure 4. Short-run loss while operating: at $Q = 15$, price is below ATC but above AVC. The shaded loss is about \$115, smaller than the \$160 fixed cost if the firm shuts down.

Read the graph. The $MR = MC$ intersection still determines 15 boxes. But $P < ATC$ shows a loss. Crucially, $P > AVC$ shows why it is better to keep producing *for now*. The short-run shutdown condition depends on AVC, not ATC.

5. Shut down in the short run; exit in the long run

This is a central distinction. A business may stop production temporarily but still owe its rent or payments on equipment. Leaving the industry permanently eventually permits it to avoid the full costs of staying in business. Confusing these decisions leads to wrong answers even when the graphs are drawn correctly.

DEFINITION Shutdown, exit, and sunk cost

A **shutdown** is a temporary short-run decision to produce zero units. The firm still pays its unavoidable fixed costs. **Exit** is a long-run decision to leave the market, allowing avoidable costs of continuing in the industry to be escaped. A **sunk cost** is a cost already incurred that cannot be recovered; it should not affect a decision about what to do next.

KEY RESULT Two rules students must distinguish

Short run: shut down if $TR < VC$, equivalently $\boxed{P < AVC}$ at the best positive output. At $P = \min AVC$, producing and shutting down give the same profit.

Long run: exit if the revenue achievable by remaining is less than total avoidable economic cost: $\boxed{P < ATC}$ at the relevant best output. Under the usual competitive cost assumptions, entry is attractive when price exceeds minimum long-run average total cost.

Worked example: The financial meaning of shutdown

A printing business would earn \$90 by staying open tomorrow and incur \$110 in extra operating costs. Its unavoidable rent is \$60 even if it closes tomorrow. If it opens, profit for the day is $90 - 110 - 60 = -\$80$. If it shuts for the day, profit is $-\$60$. It should shut down: running would add a \$20 loss. The \$60 fixed obligation is irrelevant to the comparison because it occurs either way.

If, instead, revenue were \$130 against the same \$110 variable costs, opening would lose \$40 overall, but closing would lose \$60. Open and reduce the loss. **This is why a firm can rationally operate at a loss.**

KEY RESULT The three price regions, expressed without calculus

For the standard U-shaped cost curves:
$P < \min AVC$: shut down now.
$\min AVC < P < \min ATC$: produce in the short run but make a loss; exit in the long run if the unfavorable conditions persist.
$P > \min ATC$: produce with positive economic profit at the optimizing output.
At $P = \min ATC$, economic profit is zero. At $P = \min AVC$, shutdown and production are tied.

Careful: the two average-cost thresholds have different roles. ATC contains fixed cost and determines whether revenues cover *all* costs. AVC excludes fixed cost and determines whether continuing output contributes anything toward those unavoidable costs.

6. The short-run supply curve

A firm does not supply whatever quantity it happens to have capacity to produce. Given the market price, it chooses its output using marginal cost, then checks whether it should open at all. We can therefore derive a supply curve from its cost curves rather than treating supply as a separate rule to memorize.

KEY RESULT Competitive firm's short-run supply

The firm's **short-run supply curve is the upward-sloping part of its MC curve at and above minimum AVC.** At lower prices, the firm supplies zero. At exactly the shutdown price it can be indifferent between zero and the positive quantity at minimum AVC.

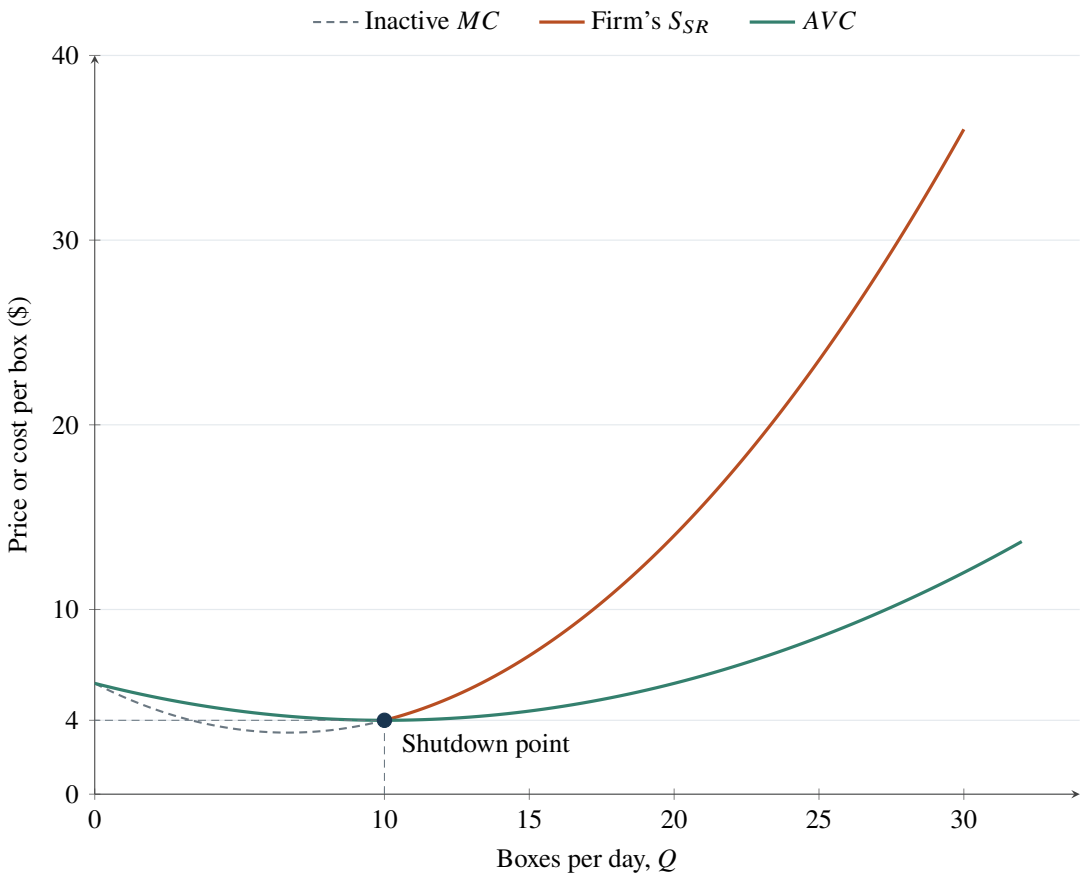


Figure 5. The firm's short-run supply is its rising MC curve beginning at minimum AVC. The dashed lower section of MC is *not* an operating supply curve.

Read the graph. The bottom of the AVC curve occurs at 10 boxes and \$4 per box. Below \$4, the firm produces zero. At a higher market price, go horizontally to the rising orange MC curve, then down to read the firm's quantity. This is how the price-taking firm responds to a change in market price.

From one firm to the industry

Market supply is the **horizontal sum** of individual firms' supply decisions: add how many boxes each firm is willing to provide at the same market price. If 60 identical firms each offer 15 boxes at a given price, the market supplies $60 \times 15 = 900$ boxes. In the short run the number of firms is usually fixed; if the price increases, existing firms move along their own supply curves and market output rises.

7. Entry and exit

The short run holds the number of firms essentially fixed. In the long run, profits and losses change the number of sellers, which changes *market supply*. The incentive is economic profit, not whether a firm's accounting statement reports positive earnings.

> **KEY RESULT How entry and exit change market outcomes**
>
> **Positive economic profits** attract entry. More sellers shift *market supply* right, lowering price and reducing existing firms' profits.
>
> **Economic losses** induce exit. Fewer sellers shift market supply left, raising price and reducing the losses of firms that remain.
>
> Under free entry, identical costs, and no cost changes as the industry grows, this process stops when surviving firms earn **zero economic profit**.

Worked example: Why positive profit does not persist in the standard model

Suppose the typical firm can make boxes at minimum long-run average total cost of $14. At a market price of $19, firms have an incentive to expand or enter. Total market supply increases, reducing price. Entry slows as profits shrink and stops when price reaches $14. If the market price instead stayed below $14, firms would leave, reducing supply and pushing price back upward.

> **DEFINITION Zero economic profit versus zero income**
>
> **Zero economic profit** means total revenue just covers all economic costs, **including** the opportunity cost of the owner's time, capital, and other resources. It does *not* mean that owners receive no income. Their normal compensation for those resources is included in economic cost.

So what? If a former employee starts her own printing business, she must count the salary she gives up as a cost. A business can report positive accounting profit yet earn zero economic profit after that forgone salary and the return on invested savings are included.

8. Firm and industry equilibrium

There are two equilibrium decisions happening at once. **The market** determines the going price using industry demand and industry supply. **Each firm** accepts that price and produces where $P = MR = MC$, if it is worth operating. In the long run, entry and exit also determine how many firms remain.

> **DEFINITION Short-run and long-run competitive equilibrium**
>
> In **short-run equilibrium**, market quantity demanded equals supplied, and operating firms choose output to maximize profits at the given market price. Economic profits or losses can occur.
>
> In the standard **long-run competitive equilibrium** with identical costs, free entry and exit, and constant input prices, firms earn zero economic profit and produce at the **efficient scale**, the output minimizing average total cost.

KEY RESULT The long-run equalities and what they mean

For the standard identical-firm model,

$$\boxed{P = MR = MC = \min ATC.}$$

$P = MR$ follows from price taking. $MR = MC$ follows from the output decision. $P = ATC$ follows from zero economic profit. $MC = ATC$ occurs at the minimum of ATC, so each firm operates at efficient scale. **All four equalities have separate economic reasons.**

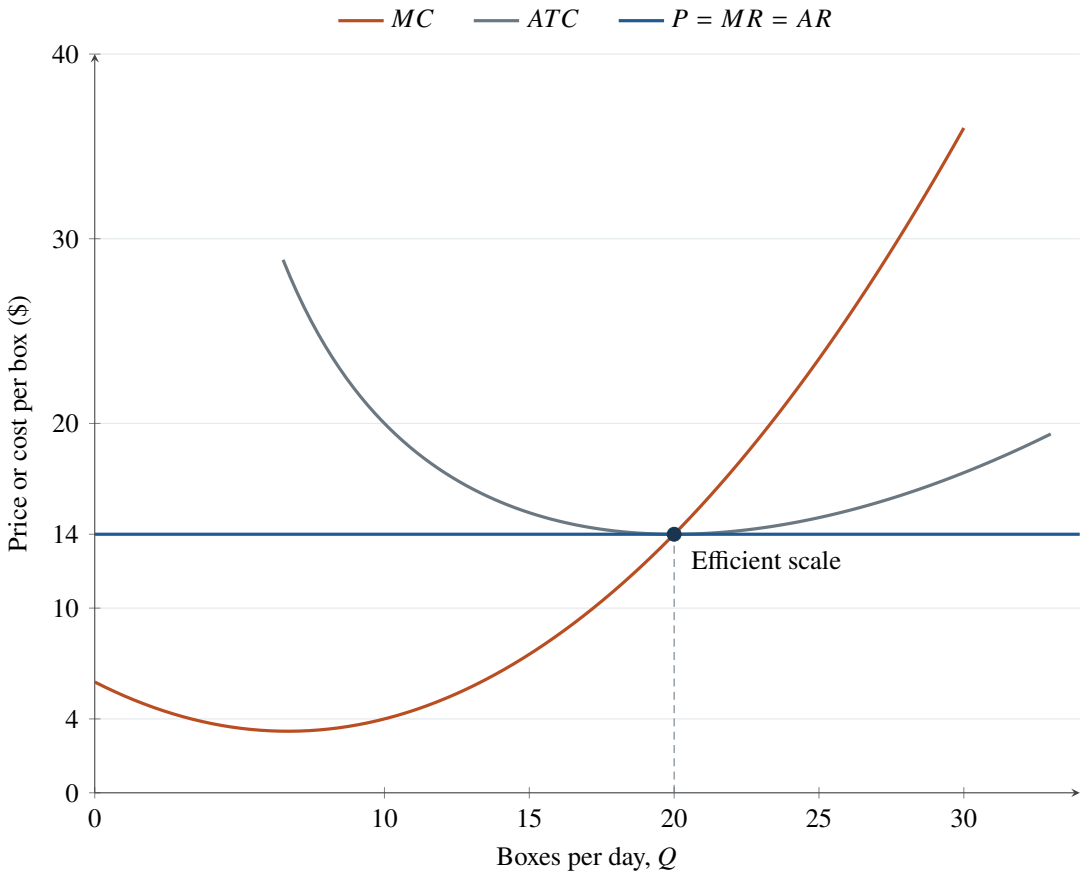


Figure 6. Standard long-run firm equilibrium. At $Q = 20$, the price-taking firm's MR equals MC, and ATC is at its minimum. Total revenue equals total economic cost.

Read the graph. The blue price line *touches* the bottom of ATC at exactly the same quantity where it crosses rising MC. Price covers average total cost, but no surplus economic profit remains to attract another entrant.

KEY RESULT Firm supply is not the same as industry supply

In the textbook's simplified long-run model, an individual firm continues only when price covers average total cost. Its positive-output long-run supply is the rising part of MC above minimum ATC. The **industry** supply curve also depends on how many firms enter and leave. Do not confuse these two curves.

How the industry adjusts to a permanent increase in demand

Begin in long-run equilibrium with price at minimum ATC. When customers want more of the product, demand shifts right. In the **short run**, with the number of sellers fixed, the market price rises and existing firms raise output, often earning positive profit. In the **long run**, that profit attracts entry. Industry supply shifts right, price falls, and entry eventually stops when economic profit returns to zero. The industry serves more customers because there are more firms, not because each identical firm permanently produces more than its efficient scale.

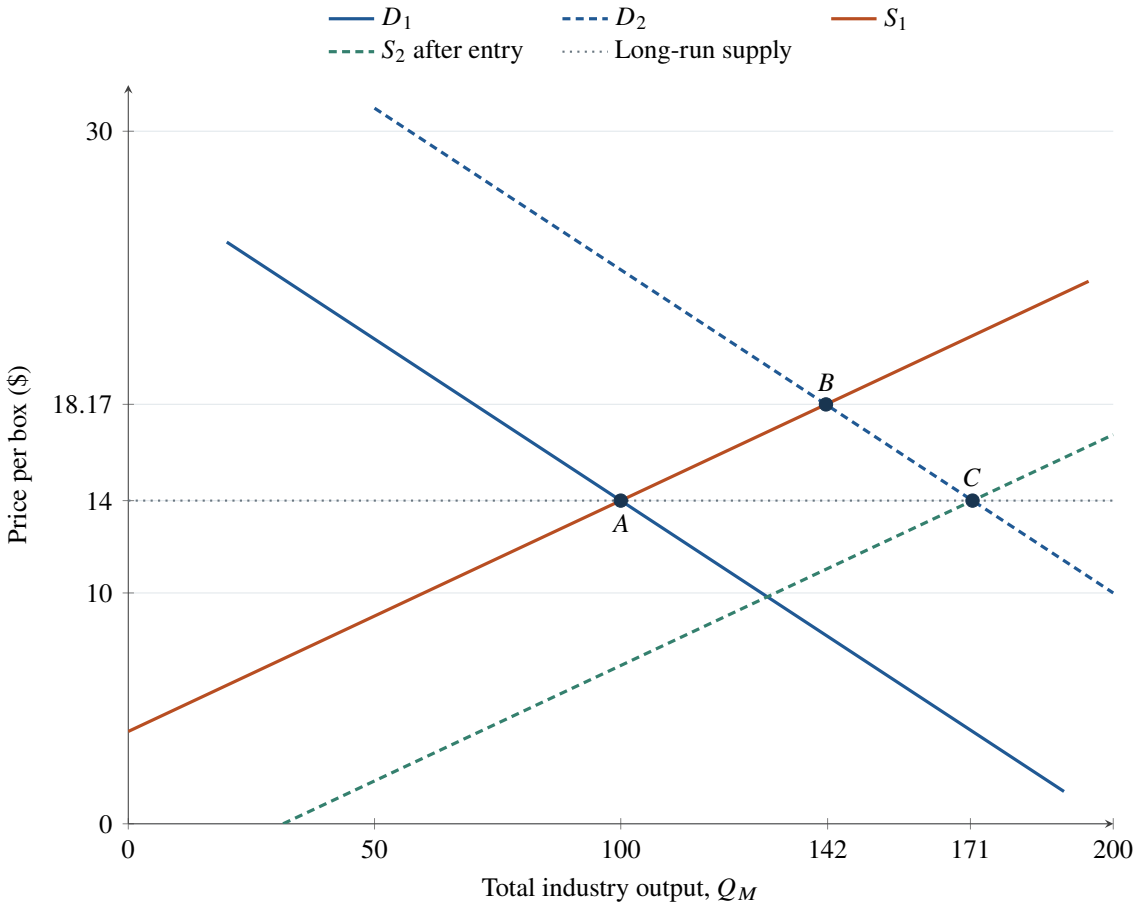


Figure 7. Demand shifts from D_1 to D_2. Initially, the market moves from A to higher-price B along S_1. Entry then shifts supply to S_2, moving the market to C at the original long-run price of \$14, the minimum of ATC.

Read the graph. Follow the story, not merely the arrows. At A firms earn zero economic profit. At B the higher price temporarily makes production more profitable. The resulting entry makes more boxes available at each price; price returns to its original level at C, while *industry quantity is larger*. The horizontal long-run industry supply result requires identical firms and no increase in input costs from industry expansion.

When is the long-run industry supply curve not horizontal?

If firms have different costs, higher-cost firms may enter only at higher prices. If a growing industry bids up a scarce input, every firm's costs may rise as more firms enter. In either case, **long-run market supply can slope upward**; the horizontal curve is not an unconditional law. Lower-cost firms may retain economic rents or profits when entry cannot replicate their cost advantage.

Why does competition matter for efficiency?

With no external costs or benefits and under the competitive model, the market price reflects the marginal value of the good to buyers. Because the firm chooses $P = MC$, the final unit produced has a value equal to its opportunity cost. Making one extra unit would cost more than it is worth; making one fewer would forgo a worthwhile trade. This explains the efficiency result, rather than merely asserting it.

Quick review: distinguish the rules

The best way to answer conceptual questions is to ask *which decision* the question describes. The output decision, the shutdown decision, and the entry-or-exit decision are not interchangeable.

Question being asked	Correct economic test and reason
How much should the firm produce?	Choose where MR meets rising MC (or compare each next unit's MR and MC in a table).
Is it earning economic profit?	At chosen Q, compare P with ATC; $\pi = (P - ATC)Q$.
Should it operate tomorrow?	Compare P with AVC. Fixed costs that cannot be avoided tomorrow do not change the choice.
Should it remain in the industry?	Compare revenue with total avoidable economic cost over the long run; persistent $P < ATC$ leads to exit.
What is the firm's short-run supply?	Rising MC above minimum AVC, not the whole MC curve.
What happens after firms earn profits?	Entry increases industry supply, reducing price and economic profit.
What characterizes standard long-run equilibrium?	With free entry, identical firms, and constant input prices: $P = MR = MC = \min ATC$.

Three common mistakes, corrected

Mistake 1: "A loss-making firm must close immediately." False. At a price of \$10, if a firm has $AVC = \$7$ and $ATC = \$13$ at its optimal output, it covers its extra operating costs and contributes \$3 per unit toward fixed cost. It may rationally operate at a loss in the short run.

Mistake 2: "When $MR = MC$, profit must be positive." False. $MR = MC$ tells the firm which positive quantity is best. Its profit depends on ATC at that quantity, and it may still be better to shut down when price is below AVC.

Mistake 3: "Zero economic profit means the owner receives nothing." False. Economic cost already includes the value of the owner's time and investment. Zero economic profit means those resources are receiving their normal opportunity-cost return.

> **KEY RESULT One final connection**
>
> The market sets the competitive price, the firm chooses its output, and entry or exit changes the number of firms. In the short run, these are different decisions with different consequences. In the long run, competition links them together through the zero-economic-profit condition.